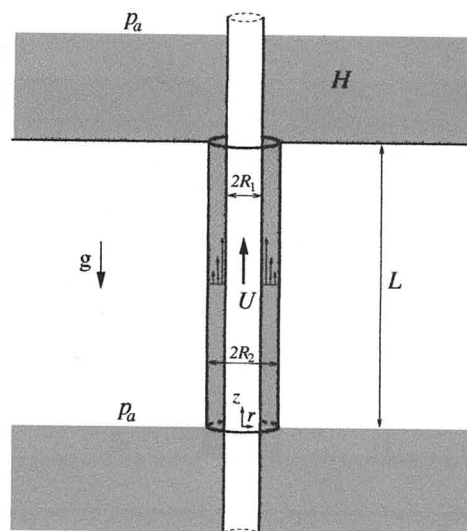


Problem 4:

To fill an open container of depth H located at a height L above the surface of a reservoir we use the viscous pumping system sketched in the figure, including a pipe of radius R_2 connecting the surface of the reservoir with the bottom of the container and an infinitely long, inner, coaxial rod of radius R_1 that moves vertically with velocity U , dragging the water into the inter-cylinder gap.

1. Give the condition for the flow in the gap to be dominated by viscosity.
2. Obtain the velocity profile $v_z(r)$ across the gap.
3. Determine the loss of reduced pressure per unit length $P_l = -\partial P / \partial z$.
4. Compute the force acting on the moving shaft and the power needed to move it.
5. Determine the volume flow rate Q of the pumping system, including the minimum velocity U_{min} required for $Q > 0$.



3) $P_l = \text{constant} = -\frac{\partial P}{\partial z} = \frac{P_a - (P_a + \rho g(H+L))}{L} = -\frac{\rho g(H+L)}{L}$

1) $\frac{\rho U (R_2 - R_1)^2}{\mu L} \ll 1$ 2) $0 = P_l + \frac{\mu}{r} \frac{d}{dr} \left(r \frac{dv_z}{dr} \right) \begin{cases} r=R_1: v_z=U \\ r=R_2: v_z=0 \end{cases} \Rightarrow v_z = \frac{P_l}{4\mu} (R_2^2 - r^2) + \left[U - \frac{P_l}{4\mu} (R_2^2 - R_1^2) \right] \frac{\ln(R_2/r)}{\ln(R_2/R_1)}$

4) $W = U 2\pi R_1 L \tau_w, \tau_w = \mu \frac{dv_z}{dr} \Big|_{r=R_1} = -\frac{P_l R_1}{2\mu} - \frac{\mu}{R_1} \left[U - \frac{P_l}{4\mu} (R_2^2 - R_1^2) \right] \frac{1}{\ln(R_2/R_1)}$

5) $Q = \int_{R_1}^{R_2} 2\pi r v_z dr = -2\pi \left[\frac{P_l}{16\mu} (R_2^4 - R_1^4) + \frac{\left(U - \frac{P_l}{4\mu} (R_2^2 - R_1^2) \right)}{4 \ln(R_2/R_1)} \left[R_2^2 (2 \ln R_2 - 1) - R_1^2 (2 \ln R_1 - 1) \right] \right]$

$U_{min} = -\frac{P_l}{4\mu} (R_2^2 - R_1^2) \left[\frac{(R_2^2 + R_1^2) \ln(R_2/R_1)}{R_2^2 (2 \ln R_2 - 1) - R_1^2 (2 \ln R_1 - 1)} - 1 \right]$